CCLVM1

A museum-quality VM/CMS environment

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Classical Computing Laboratory at IBM Poughkeepsie
Poughkeepsie, NY

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- CCLVM1 is a museum-quality VM/CMS Environment hosted in the Classical Computing Laboratory at IBM Poughkeepsie
  - An employee-run historical IT center on the IBM campus in Poughkeepsie, NY
  - Open to all IBM employees and official site visitors by appointment
  - Currently undergoing renovations, expected reopening in 2028
- Accurately recreates the kind of timesharing environments used at IBM and other companies from the 1970s through the 1990s

- IBM 3006-B01 System/390 Integrated server
- IBM 3174-21L Establishment Controller (channel-attached)
- IBM 3278-2 Display Station
- IBM 3278-5 Display Station
- IBM 3279-S2B Color Display Station
- IBM 3279-S3G Color Display Station
- IBM 3299 8-Port Coax Multiplexor
- IBM 4224-1 Dot Matrix Line Printer

IBM 3006-B01 System/390 Integrated Server CCLVM1

- Introduced on November 12, 1998
- Enterprise-grade Pentium II server with mainframe co-processor
- Based off of "Personal System/370" and "Personal System/390" lines of mainframe-compatible PC workstations
- Has an OS/2 host operating system on the Pentium II, running S/390 operating systems in parallel on the mainframe co-processor
  - OS/2 host conceptually serves the same function as the S/390 Support Element, just with a completely different software stack
- Allowed for mainframe functionality on real hardware, without the need for expensive additional S/390 devices such as 3480/90 tape drives or 3390 disk storage
- Succeeded by the IBM 7060 Multiprise 3000, a real mainframe in same form-factor case

- Introduced in 1986
- Needed to connect terminals and printers to mainframes
- Processes I/O operations for the terminals and printers
- Ours is a rack-mount model that connects with System/360-style "bus and tag" channel cables
  - Large floor models and small desktop models were also available
- The 3174 could be expended with a range of options and adapters: channel, ESCON, serial (V.24/V.35), coax, fiber optic "coax", ISDN, token-ring, ethernet

- Monochrome mainframe terminal introduced in 1977

- Introduced the following features:
  - Extended Highlighting: ability to set highlighting attributes on individual characters as well as on fields (blinking, character set, reverse video, and underscored)
  - Accented characters and dead keys
  - Programmed Symbols: programmable characters, able to display monochrome graphics (with 3620 Character Set Extension and 5790 Programmed Symbols)

- CCLVM1 features two models:
  - Model 2: 80 x 24 character resolution
  - Model 5: 132 x 27 character resolution, switchable to 80 x 24

- First IBM color terminal, introduced in 1979
- Uses same "Extended Data Stream" as the 3278
  - Think of it as a 3278 with color
- Two color sets available:
  - Base color: Greed, Red, Blue, White
  - Extended color: Base colors plus Pink, Yellow, and Turquoise
- CCLVM1 features two models:
  - Model S2B: 80 x 24 character resolution with extended color
  - Model S3G: 80 x 32 character resolution with extended color and programmed symbol set graphics

- Low-end enterprise line printer from 1986
- Speed of around 200 characters-per-second in draft mode
  - Compare with high-end 6262-D22 of same era: 2,200 lines per minute!
- Several variants were available:
  - System/36, System/38, and AS/400 (twainax interface)
  - System/370 and System/390 (coax interface)
  - Series/1 (parallel ASCII interface)
  - System/88 (serial ASCII interface)

- VM is a ***hypervisor***, an operating system that serves virtual machines
  - The current version, z/VM, is used to host large Linux server farms
- Introduced internally in 1967 as **CP-67**, first sold to the public as **VM/370** in 1972.
- Was conceived as a solution to the ***time-sharing problem***, that is, the problem of allowing more than one person to use a computer simultaneously
  - Each user would get their own virtual machine, which could run any System/360 or System/370 operating system
  - General purpose users used **CMS**, a single-tasking operating system with a non-hierarchical filesystem
  - Other popular uses were for operating system development, and for partitioning a System/370 for running multiple instances of VSE

- By the 1980s, IBM had extensive internal infrastructure built around VM and CMS
  - In a world before PCs, IBMers used CMS virtual machines for tasks like email, calendaring, document preparation, application development, and end-user applications
  - A global CMS-based bulletin board system served as the corporate communications and file sharing backbone for IBM

- As a result, VM and CMS was the host platform for many IBM innovations:
  - **Generalized Markup Language (GML)**, a set of markup tags that are the direct ancestor to XML
  - **IBM System R**, the first implementation of SQL
  - **LEXX**, the first text editor to use live parsing and colored syntax highlighting
  - **PROFS**, the first commercially-successful office automation software
  - **Rexx**, a high-level programming language considered a precursor to Tcl and Python
  - The first RISC compiler and linker were written on YMS, a research operating system running as a VM/370 guest

- CCLVM1 is running **VM/ESA 2.4.0**, a version of VM from 1999
- This can be considered the last version of "old world" VM
  - In December of 1999, IBM would formally add S/390 support to the Linux kernel. Linux would eventually become the dominant guest operating system on VM and use of VM/CMS as an end-user timesharing environment would greatly diminish.
  - In 2000 the product would be renamed z/VM with the introduction of 64-bit IBM z/Architecture support
- Rationale for running VM/ESA 2.4.0 on the IBM 3006:
  - Current during the period in which the S/390 Integrated Server was a supported product
  - Improved reliability over VM/ESA 2.3.0
  - S/390 processor in the 3006 uses 31-bit addressing, so no need for the z/Architecture support provided by z/VM 3.1

- Great pains have been taken to outfit CCLVM1 with a wide array of software appropriate for a 1999-era VM/CMS timesharing system:

Compilers	Available for BASIC, COBOL, Fortran, PL/I, and Rexx
DCF	Document Composition Facility, used for markdown-based document publishing
Demos and Games	Curated by current and former IBMers, in addition to examples found in internet archives such as "the CBT tape"
OfficeVision/VM	Integrated office suite with mail, calendaring, and other features. Successor to PROFS.
TOOLS	Filesharing and bulletin board system

- We developed a new CMS application called **CCLMENU** to help visitors to the Classical Computing Laboratory get the most out of CCLVM1

- Design philosophy

  - Users with no prior VM or CMS experience should be able to accomplish meaningful tasks with minimal assistance

  - UX should be visually appealing and show off the qualities of IBM 3270-series terminals

  - The application should take advantage of user libraries developed by earlier generations of IBMers ("standing on the shoulders of giants")

- To that end, we used two libraries:

  1. **Display Input/Output Facility v.3 (DIOF-3)**, also known as "IOS3270", is a library that allows CMS Rexx programmers to write full-screen panel applications without manipulating the 3270 datastream themselves

  2. **NEATWIND** creates and manages overlapping pop-up windows and dialog boxes, with the ability to use graphical icons on terminals that support it (such as our 3279-S3G)

- **In this manner, we guide the user through a curated selection of** CMS applications without them needing to know how to use CMS itself

  – Application panels ask simple numeric response or yes-no questions

1. Create a Banner, Sign, or Calendar

   – Pop-up windows guide users through the process of designing a multi-page vertical banner or a single-page sign. An IBM VS FORTRAN version of the famous Snoopy calendar generator can also be used to create calendars for arbitrary month/year ranges

2. Play a Game

   – A selection of over fifteen games is presented, taken from sources such as ***Creative Computing*** magazine, David H. Ahl programming books, the CBT tape archive, and the personal collections of IBM retirees

3. View Cool Demos

   – Similar to the above, but with interesting demos

4. Talk with our Advanced AI Chatbot

   – A port of the ***Creative Computing*** version of ELIZA, with the option to print transcripts

Special Thanks

This hard work would have been impossible without the generous time, donations, and guidance of a dedicated community of VM/CMS systems programmers and developers:

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Foils 5.3 .	CCLVM1 - 18 foils - 0 notes
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Fonts Format	MIX Foils
Options	Pitch 6 10 - Bind 7 7 Tags . . .
DCF	Release 4.0.0